Chapter 19

A General Understanding of Metrology

The field of metrology spans a multitude of different disciplines. Metrology incorporates an ensemble of knowledge gathered from multiple diverse fields such as mathematics, statistics, physics, quality, chemistry, mechanical, and computer science; all applied with a liberal sprinkling of common sense. Essential to the field of metrology is understanding the fundamental methods by which objects and phenomena occur and are measured, as well as the means for assigning values to measurements and the uncertainty of these assigned values. Encompassing some of these essentials are the establishment and maintenance of units, measurement methods, measurement systems, measurement capability, measurement data, measurement equipment specifications, measurement standards usage, measurement confidence programs, etc. The adage "A chain is only as strong as its weakest link" also applies to metrology because metrology essentials are interdependent; each relies on an assortment of clearly stated definitions and postulations. The ensemble of metrology essentials lays the foundation for the realization and agreement throughout the world that measurements are accepted and appropriate for their intended purposes. Mathematics is widely considered the universal language; the metrology essentials extend that language into our daily existence as quantifiable, attributed information without which our world, as we know it, could not exist.

The basic concepts and principles of metrology were formulated from the need to measure and compare a known value or quantity to an unknown to define the unknown relative to the known. This may seem like double-talk, but upon further investigation you can see that what is being described is a method for determining the value of an unknown by assigning it a quantity of divisions commonly referred to as units (for example, meters, degrees Celsius, minutes, and so on). Everything we buy, sell, consume, or produce can be compared, measured, and defined in terms of units of a measurement. Without commonly agreed-on units, it would not be possible to accurately quantify the passing of time, the length of an object, or the temperature of one's surroundings. In fact, almost every aspect of our physical world can be related in terms of units of measurement. Units allow us to count things in a building-block type fashion, so they have meaning beyond a simple descriptive comparison such as smaller than, brighter than, longer than, and so on. Determination of measurement units that are deemed acceptable and repeatable, and maintaining them as measurement standards, lies at the heart of fundamental metrology concepts and principles.